

SCS 1307
Probability & Statistics

by
Dr Dilanthi Gamage Dona
Department of Statistics
University of Colombo

Random Variables

- Many real-world experiments produce uncertain outcomes. To analyse them mathematically, we often convert outcomes into numbers.
Hence we talk about random variables.

- **Random variables**

Let Ω be the outcome space corresponding to a random experiment.
A function $X: \Omega \rightarrow \mathbb{R}$ is called a random variable

Discrete Random Variables

The random variable X is a discrete random variable if the values taken by the random variable contains only a finite number of sample points.

Example:

A coin is tossed three times, and the sequence of Heads and Tails is observed.

The outcome space is

$$\Omega = \{hhh, hht, hth, thh, htt, tht, tth, ttt\}$$

Let X = number of Heads observed,
 Y = number of Tails observed,
 $Z = X \cdot Y$.

The functions X, Y and Z are conveniently specified by the table below.

	HHH	HHT	HTH	THH	HTT	THT	TTH	TTT
X								
Y								
Z								

Therefore $\Omega_X = \Omega_Y = \{?\}$ and $\Omega_Z = \{?\}$.

Exercise

A coin is tossed until two successive tosses give the same result.

Let X = number of Heads observed and Y = total number of tosses made.

Give Ω_X , the sample space of X and Ω_Y , the sample space of Y .

Solution

$$\Omega = \{.....\}$$

$$\Omega_X = \{.....\}$$

$$\Omega_Y = \{.....\}$$

Probability Distributions

Let X be a discrete random variable defined on the outcome space Ω . Let x be a value in Ω_X , the sample space of X . We are interested in the event ' X takes the value x '. We are also interested in the probability associated with this event.

Definition

Let X be a random variable with sample space Ω_X . The probability function of X is the function P_x defined by ,

$$P_x = \left\{ \begin{array}{ll} P(X=x) & \text{if } x \in \Omega_X; \\ 0 & \text{Otherwise} \end{array} \right\}$$

By the probability distribution of X we mean the way in which the probability mass is distributed over the values in Ω_X .

The probability function P_x is one way of specifying the probability distribution of X .

Example

Toss a coin three times and observe the sequence of Heads and tails.

Here $\Omega = \{hhh, hht, hth, thh, htt, tht, tth, ttt\}$.

Suppose the results on the tosses of the coin are independent, and that any given toss falls Heads with probability $1/3$.

Define X = Number of Heads observed and Y = number of Tails observed.

Solution

$\Omega = \{hhh, hht, hth, thh, htt, tht, tth, ttt\}$.

X = Number of Heads observed and Y = number of Tails observed.

- Probabilities can be specified as follows.

ω	HHH	HHT	HTH	THH	HTT	THT	TTH	TTT
$P(\omega)$								
$X(\omega)$								
$Y(\omega)$								

Note that $P_x(0) = ?$, $P_x(1) = ?$, $P_y(2) = ?$.

Functions of Random Variables

It is sometimes useful to define random variables as functions of other random variables.

Examples : If X is a random variable, then so are $2X+5$, and X^2 .

Example

A machine is used to cut rectangular shapes out of sheets of metal. Switch A has 2 settings which are used to select the length L of the rectangle (2cm or 3cm). Switch B has three settings which are used to select the width W of the rectangle (1cm, 2cm or 3cm). A mechanic tests the machine by selecting settings for switches A and B and making a rectangle. The length and width are observed. Suppose the settings are chosen randomly in a way so that in the Table below.

	(2,1)	(2,2)	(2,3)	(3,1)	(3,2)	(3,3)
$P(\omega)$	0.12	0.13	0.25	0.20	0.15	0.15

Example

- a. Find p_L , the probability distribution of length
- b. Find p_W , the probability distribution of width
- c. Find p_X , the probability distribution of perimeter $X=2L+2W$

	(2,1)	(2,2)	(2,3)	(3,1)	(3,2)	(3,3)
$P(\omega)$	0.12	0.13	0.25	0.20	0.15	0.15

Solution

a. Find p_L , the probability distribution of length

	(2,1)	(2,2)	(2,3)	(3,1)	(3,2)	(3,3)
$P(\omega)$	0.12	0.13	0.25	0.20	0.15	0.15

Probability Distribution of length

L		
$P(L)$		

Solution

b. Find p_W , the probability distribution of width

	(2,1)	(2,2)	(2,3)	(3,1)	(3,2)	(3,3)
W						
$P(\omega)$	0.12	0.13	0.25	0.20	0.15	0.15

Probability Distribution of width

W			
$P(W)$			

Solution

c. Find p_X , the probability distribution of perimeter

	(2,1)	(2,2)	(2,3)	(3,1)	(3,2)	(3,3)
X						
$P(\omega)$	0.12	0.13	0.25	0.20	0.15	0.15

Probability Distribution of perimeter

X				
$P(X)$				

Expectation and Variance

When the range Ω_X of a random variable $X: \Omega \rightarrow \mathbb{R}$ contains only a few points it is easy to 'absorb' the entire probability distribution.

However, it is often the case that Ω_X is very large.

In these cases it is useful to talk about *expectation* that can help to provide a measure of 'location' and a measure of 'spread'

Definition of Expectation

The *expectation* of a random variable X defined on an outcome space Ω is the value $E(X)$ given by

$$E(X) = \sum_{\omega \in \Omega} X(\omega)P(\{\omega\})$$

This quantity is also referred to as the *mean* of X , or the *mean of the probability distribution* of X or the *expected value* of X .

Consider the following probability model:

(ω)	ω_1	ω_2	ω_3	ω_4	ω_5	ω_6
$P(\{\omega\})$	0.1	0.2	0.2	0.1	0.3	0.1
$X(\omega)$	3	3	2	1	3	2

Calculate the expected value of X .

$E(X)$?

Properties of Expectation

1. $E(X) = \sum_{X \in \Omega} X P(x)$
2. $E(aX + b) = a E(X) + b$ where a & b are constants
3. Let X and Y be random variables defined on the same outcome space Ω .

$$E(X + Y) = E(X) + E(Y)$$

Exercise

The random variable X has probability distribution given by

X	-2	-1	0	1	2
$P(X)$	0.3	0.1	0.2	0.1	0.3

- (a) Find $E(X)$
- (b) Find $E(X^2)$

Solution

The random variable X has probability distribution given by

X	-2	-1	0	1	2
X^2					
$P(X)$	0.3	0.1	0.2	0.1	0.3

(a) $E(X) =$

(b) Find $E(X^2) =$

Variance

The mean of the random variable X is the value of $E(X)$. The variance of X is the quantity

$$\sigma_x^2 = E[(X - \mu)^2] = \sum_{x \in \Omega_X} (X - \mu)^2 P(x).$$

Then σ_x is called the *standard deviation* of X . Both the standard deviation and the variance can be used as measures of the spread of a probability distribution, however the standard deviation is commonly used since it is a measurement in the same unit as X .

Properties of Variance

Property 1: $V(X) = E(X^2) - \mu^2$

Property 2: If c is any constant and X is a random variable then

$$\text{Var}(cX) = c^2 \text{Var}(X)$$

Property 3:

If X and Y are independent random variables,

$$\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) \text{ and}$$

$$\text{Var}(X-Y) = \text{Var}(X) + \text{Var}(Y)$$

Exercise

Toss an unbiased coin once and win $\$k$ if it is a Head and lose $\$k$ if it is a Tail. Find the standard deviation of the profit X if the probability distribution of X is given by

X	$-k$	K
$P(X)$	0.5	0.5

Solution

X	$-k$	k
X^2		
$P(X)$	0.5	0.5

$$E(X) = \mu = 0$$

$$V(X) = V(X) = ?$$

$$\sigma_x = ?$$

Exercise

Suppose that a game is played with a single die assumed fair. In this game a player wins \$20 if a 2 turns up, \$40 if a 4 turns up; loses \$30 if a 6 turns up; while he neither wins or loses if any other face turns up. Find the expected sum of money won.